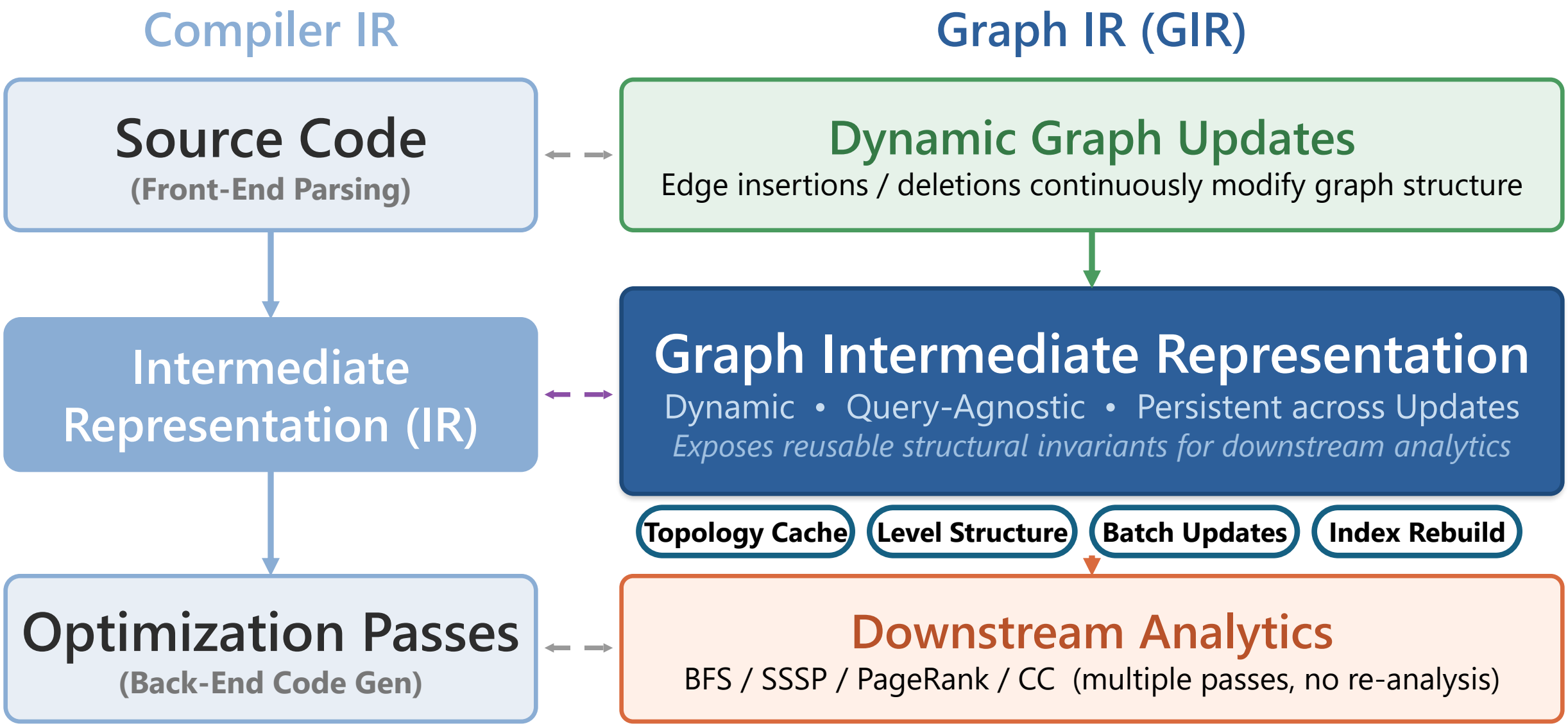

Compiler IR
Graph IR (GIR)
Source Code
(Front-End Parsing)
Dynamic Graph Updates
Edge insertions / deletions continuously modify graph structure
Intermediate Representation (IR)
Graph Intermediate Representation
Dynamic • Query-Agnostic • Persistent across Updates
Exposes reusable structural invariants for downstream analytics
Topology Cache
Level Structure
Batch Updates
Index Rebuild
Optimization Passes
(Back-End Code Gen)
Downstream Analytics
BFS / SSSP / PageRank / CC (multiple passes, no re-analysis)